National University
of Computer and Emerging Sciences
Chiniot-Faisalabad Campus

EE1005 – Digital Logic Design

Assignment Number 3

Fall 2022

Maximum Marks: 50 **Due Date:** 01 November, 2022

Submitted By

Name: ____________________________

Student ID: ____________________________

Section: ____________________________

Submitted To

Dr. Adil Zulfiqar
Assistant Professor & In-charge – EE Department

Submission Date

___ November 2022

Questions	1	2	3	4	5	Total marks
Marks	10	5	15	10	10	50
Obtained Marks						

Question No. 1: **[05 Marks]**
Design a combinational logic circuit that generates the 9's complement of BCD digits.

Question No.2: **[10 Marks]**
Design a full-subtractor circuit with three input x, y, Bin and two output Diff and Bout. The circuit subtracts x-y-Bin, where Bin is the input borrow, Bout is the output borrow, and Diff is the difference. Implement the full subtractor using NAND gates only.

Question No. 3: **[10 Marks]**
Design a combinational logic ciruit that implements a 4-bit BCD to Excess-3 Code convertor:

Question No. 4: **[15 Marks]**
Implement the following function using Decoders that consist of NAND gates only:

a) $F_1 = x^/yz^/ + xz$
b) $F_2 = xy^/z^/ + x^/y$
c) $F_3 = x^/y^/z^/ + xy$

Question No. 5: **[10 Marks]**
Implement the following Boolean Functions with a 16×1 multiplexer:

a) $F(A,B,C,D) = \sum(0,2,5,8,10,14)$
b) $F(A,B,C,D) = \prod(2,6,11)$